

- The sum of two positive quantities is equal to $2n$. The probability that their product is not less than $3/4$ times their greatest product is
(a) $3/4$ (b) $1/2$
(c) $1/4$ (d) None
- If A and B are independent events and $P(C) = 0$, then
(a) A and C are independent
(b) B and C are independent
(c) A, B and C are independent
(d) all of these
- If in a lottery there are 5 prizes and 20 blanks, then the probability of getting prize is
(a) $1/5$ (b) $2/5$ (c) $4/5$ (d) None
- Twelve balls are distributed among three boxes. The probability that the first box contains 3 balls is
(a) $\frac{110}{9} \left(\frac{2}{3}\right)^{10}$ (b) $\frac{9}{110} \left(\frac{2}{3}\right)^{10}$
(c) $\left(\frac{{}^{12}C_3}{12^3}\right) 2^9$ (d) $\frac{{}^{12}C_3}{3^{12}}$
- A card is drawn at random from a well shuffled pack of 52 cards. The probability of getting a two of heart or diamond is :
(a) $1/7$ (b) $2/7$ (c) $3/7$ (d) None
- Two non-negative integers are chosen at random. The probability that their sum is divisible by 5 is
(a) $7/25$ (b) $8/25$ (c) $9/25$ (d) $2/5$
- If the papers of 4 students can be checked by any one of the seven teachers, then the probability that all the four papers are checked by exactly two teachers is
(a) $2/7$ (b) $12/49$ (c) $32/343$ (d) None
- A six faced dice is a biased one. It is thrice more likely to show an odd number than show an even number. It is thrown twice. The probability that the sum of the numbers in the two throws is even, is
(a) $5/9$ (b) $5/8$ (c) $1/2$ (d) None
- An ordinary cube has four blank faces one face marked 2 and another marked 3. Then the probability of obtaining 9 in 5 throws is
(a) $\frac{31}{7776}$ (b) $\frac{5}{2592}$
(c) $\frac{5}{1944}$ (d) $\frac{5}{1296}$
- A and B each toss a fair coin simultaneously for 50 times. The probability that both of them will not get tail at the same toss is
(a) $\left(\frac{3}{4}\right)^{50}$ (b) $\left(\frac{2}{7}\right)^{50}$ (c) $\left(\frac{1}{8}\right)^{50}$ (d) $\left(\frac{7}{4}\right)^{50}$
- If P is chosen at random in the interval $0 \leq p \leq 5$, the probability that the roots of the equation $x^2 + px + \frac{p}{4} + \frac{1}{2} = 0$ are real, is
(a) $1/5$ (b) $2/5$ (c) $3/5$ (d) $4/5$
- Two dice are rolled one after the other. The probability that the number on the first is smaller than the number on the second is
(a) $1/2$ (b) $7/18$ (c) $3/4$ (d) $5/12$
- Three numbers are chosen from 1 to 30. The probability that they are not consecutive is ;
(a) $\frac{144}{145}$ (b) $\frac{143}{145}$
(c) $\frac{142}{145}$ (d) None
- An integer is chosen at random from the first 200 positive integers. The probability that the integers chosen is divisible by 6 or 8 is
(a) $1/3$ (b) $1/4$ (c) $1/5$ (d) None
- If a dice is thrown twice, then the probability of getting 1 in the first throw only is
(a) $1/36$ (b) $3/36$ (c) $5/36$ (d) $1/16$
- The chance to fail in physics are 20% and the chance to fail in mathematics are 10%. what are the chance to fail in at least one subject.
(a) 28% (b) 38% (c) 72% (d) 82%
- If the letters of the word 'REGULATIONS' be arranged at random, the probability that there will be exactly 4 letters between R and E is
(a) $6/55$ (b) $3/55$
(c) $49/55$ (d) None
- Out of 40 consecutive integers, two are chosen at random, the probability that their sum is odd, is
(a) $14/29$ (b) $20/39$ (c) $1/2$ (d) None
- In a class of 125 students 70 passed in mathematics, 55 in statistics and 30 in both. The probability that a student selected at random from the class has passed in only one subject is
(a) $13/25$ (b) $3/25$ (c) $17/25$ (d) $8/25$
- If two events A and B are such that $P(A^c) = 0.3$, $P(B) = 0.4$
 $P(A \cap B^c) = 0.5$, then $P\left(\frac{B}{A} \cup B^c\right) =$
(a) 0.20 (b) 0.25 (c) 0.30 (d) 0.35
- A matrix is chosen at random from the set of all 2×2 matrices with elements 0 and 1 only. The probability that the values of the determinant of the matrix chosen is positive, is
(a) $1/2$ (b) $3/16$ (c) $11/16$ (d) $13/16$

22. Three mangoes and 3 apples are in a box. If 2 fruits are chosen at random, the probability that one is a mango and the other is an apples is
(a) $3/5$ (b) $5/6$ (c) $1/36$ (d) None
23. A bag contains 3 white and 7 red balls. If a ball is drawn at random, then what is the probability that the drawn ball is either white or red.
(a) 0 (b) $3/10$ (c) $7/10$ (d) $10/10$
24. A box contains 10 good articles and 6 with defects. One article is chosen at random. What is the probability that it is either good or has a defect.
(a) $24/64$ (b) $40/64$ (c) $49/64$ (d) $64/64$
25. An almirah stores 5 black and 4 white socks well mixed. A boy pulls out 2 socks at random. The probability that 2 are of the same colour is
(a) $4/9$ (b) $5/8$ (c) $5/9$ (d) $7/12$
26. Seven digits from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9, are written in a random order. The probability that this seven digit number is divisible by 9 is
(a) $2/9$ (b) $1/5$ (c) $1/3$ (d) $1/9$
27. A bag contains an assortment of blue and red balls. If two balls are drawn at random, the probability of drawing two red balls is five times the probability of drawing two blue balls. furthermore, the probability of drawing one ball of each colour is Six time the the probability of drawing two blue-balls. The number of red and blue balls in the bag is
(a) 6, 3 (b) 3, 6 (c) 2, 3 (d) None
28. A man known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. The probability that it is actually a six is
(a) $3/8$ (b) $1/5$ (c) $3/4$ (d) None
29. If A and B are independent events such that $P(A) > 0$, $P(B) > 0$ then
(a) A and B are mutually exclusive
(b) A and $\bar{B}$ are dependent
(c) $\bar{A}$ and B are dependent
(d) $P\left(\frac{A}{B}\right) + P\left(\frac{\bar{A}}{B}\right) = 1$
30. A bag contains m white and n black balls. Two player A and B alternately draw a ball from the bag, replacing the ball each time after draw. A begins the game. If the probability of A winning (that is drawing a white ball) is twice the probability to B winning, then the ratio m:n is equal to
(a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) None

Rough Works

SCORE SHEET

a b c d	a b c d	a b c d	a b c d	a b c d	a b c d
1. ☐ ☐ ☐ ☐	6. ☐ ☐ ☐ ☐	11. ☐ ☐ ☐ ☐	16. ☐ ☐ ☐ ☐	21. ☐ ☐ ☐ ☐	26. ☐ ☐ ☐ ☐
2. ☐ ☐ ☐ ☐	7. ☐ ☐ ☐ ☐	12. ☐ ☐ ☐ ☐	17. ☐ ☐ ☐ ☐	22. ☐ ☐ ☐ ☐	27. ☐ ☐ ☐ ☐
3. ☐ ☐ ☐ ☐	8. ☐ ☐ ☐ ☐	13. ☐ ☐ ☐ ☐	18. ☐ ☐ ☐ ☐	23. ☐ ☐ ☐ ☐	28. ☐ ☐ ☐ ☐
4. ☐ ☐ ☐ ☐	9. ☐ ☐ ☐ ☐	14. ☐ ☐ ☐ ☐	19. ☐ ☐ ☐ ☐	24. ☐ ☐ ☐ ☐	29. ☐ ☐ ☐ ☐
5. ☐ ☐ ☐ ☐	10. ☐ ☐ ☐ ☐	15. ☐ ☐ ☐ ☐	20. ☐ ☐ ☐ ☐	25. ☐ ☐ ☐ ☐	30. ☐ ☐ ☐ ☐